TechBiz Security
Securing the Digital Future
Penetration Testing · Threat Intel · Academy

Week 02 Task-05

Networking Tools and Subnetting Basics

CandidateName: Asad Ali
Reference Number: TBS-INT-2026-445

Submission Date: 27|8|2026
Submitted To: TechBiz Security

Table of Contents

1) Executive Summary

Networking fundamentals play a key role in daily life of a cybersecurity professionals, as understanding how systems communicate is essential for performing security assessments and troubleshooting network-related issues. This task focused on practicing essential networking commands such as ping, traceroute, dig, and nslookup to understand practical networking concepts. Subnetting is also practiced to understand the importance of subnetting how segmenation is done in computer networks for a security purpose what does a CIDR (**Classeless inter domain routing**) notation represents in subnetting.

2) Scope

The scope of this task is to learn about networking protocols such as network connectivity how http and dns works how subnetting is used for better network utilization and securty purpose.

3) Tools used

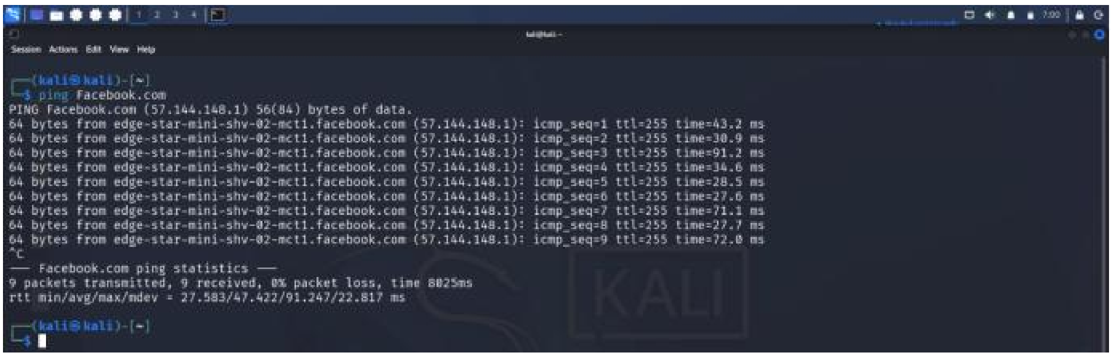
- Kali Linux
- TryHackMe room DNS in Detail
- TryHackMe room HTTP in Detail

4) Ping Command Practical Execution

4.1) Purpose of ping command

The Purpose of ping command is use to check the network connectivity. It is also use to determine if target system is alive or not.

4.2) Ouput of ping command



The ping command successfully confirm that network connectivity is working properly shown in above Image because there is no packet loss all packets that send received also. Ping command uses the icmp (**Internet Control Message Protocol**) echo requests packets to the destination server and wait for reply. IP address after resolving dns name Facebook the IP is (**57.144.148.1**)

Output Analysis

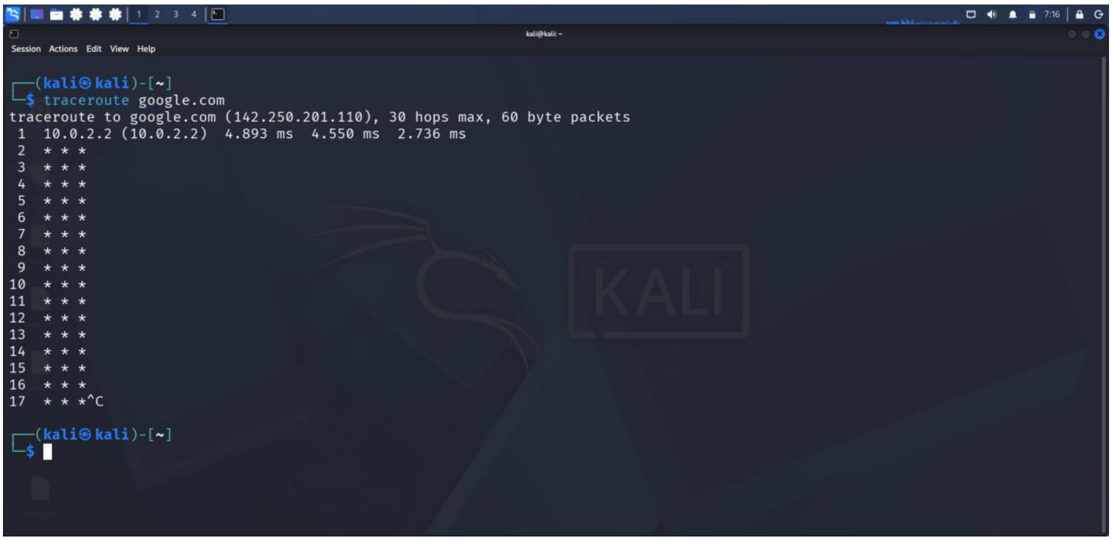
- **Size of ICMP response** = 64 bytes
- **time to live value (ttl)** = 255
- **icmp_seq** = each number of icmp request is visible in a sequence
- **Time** = 8025ms

5) Traceroute Command Practical Execution

5.1) Purpose of Traceroute command

Traceroute is a network diagnostic command used to identify the path taken by data packets from source to destination server. It shows intermediate router hops through which packet travels and measure the latency for each hop.

5.2) Output of Traceroute command



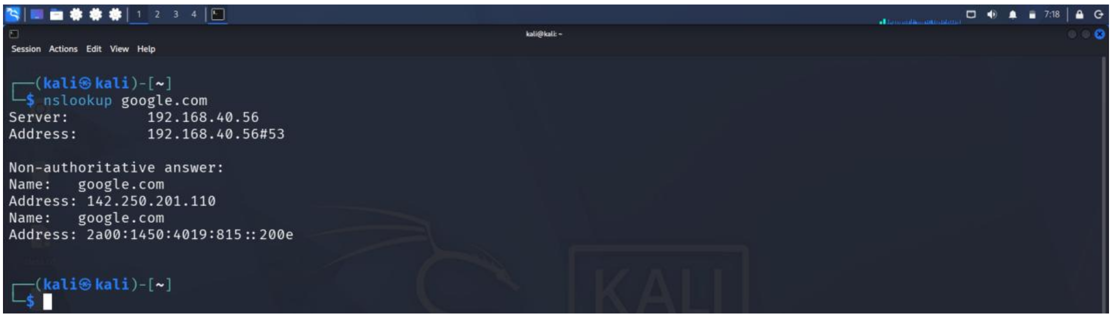
Traceroute command successfully resolved google domain name into ip address. **142.250.201.110**. In the above figure it is seen that first network hop is successfully received. Router didn't not response to other traceroute request. All other traceroute requests gets blocked because many network devices block ICMP /UDP traceroute for security reasons and don't reveal actual information.

6) DNS Lookup using nslookup Practical Execution

6.1) Purpose of dns lookup using nslookup

The main Purpose of dns lookup using nslookup (Name Server Lookup) is to query DNS servers to retreive mapping information between domain name and ip addresses.

6.2) Output of dns lookup using nslookup



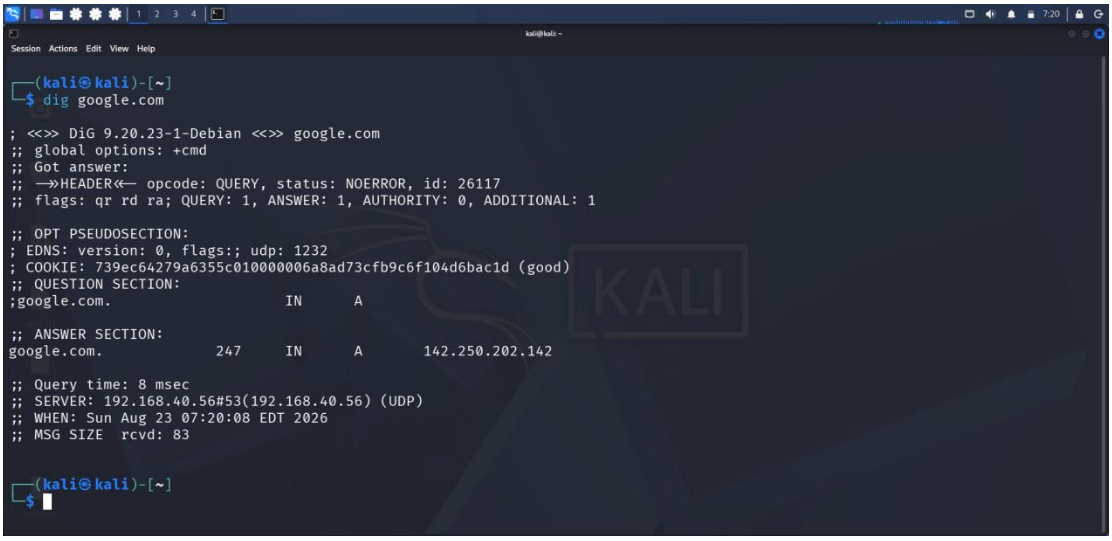
In the above Figure it is clearly shown that name google.com is successfully resolved into ip address. Both ipv4 (**142.250.201.110**) and ipv6 (**2a00:1450:4019:815::200e**) address of google server are available. **192.168.40.56** this is the local dns server answering my request.

7) DNS Lookup using dig Practical Execution

7.1) Purpose of dns lookup using dig

dig (Domain information gropper) is a tool use for dns lookup . It provides detailed raw network information about domain records. As compare to nslookup dig is the modern tool used by administrators and security professionals, It is useful in reconnaisance phase of ethical hacking by discovering hidden subdomains, mailservers and zone transfer vulnerabilities of organizations

7.2) Output of dns lookup using dig



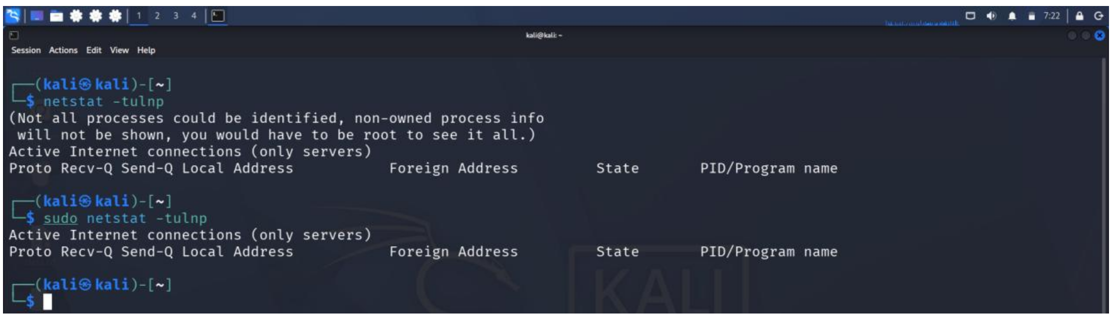
In the above picture it is clearly seen that domain name google is resolved into ip address of google using dig tool. The IP address of google server is 142.250.202.142. The header section indicats the status no error that means target domain and exist and successfully resolved into IP address. The total packet size was 83 bytes. Query time 8msec.

8) Netstat Command Practical Execution

8.1) Purpose of netstat command

The netstat command (**Network statics**) is use to display active network connections, listening ports, and services running on the network. This command is highly useful in network monitoring for administrative purpose and identifying available services during security assessment.

8.2) Output of netstat command



It is clearly shown in above Image that **netstat -tulnp** command is executed with root and without root privileges. It is found that there is no active listening ports and services were detected.

tulnp breakdown

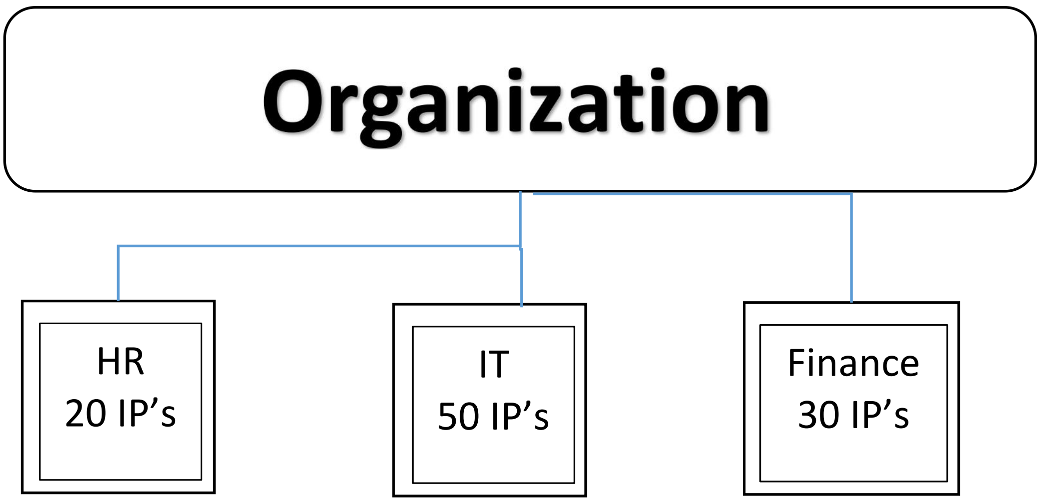
-t = tcp
-u = udp
-l = show listening ports
-n = show numerical addresses instead of resolving names.
-p = Display process/program using port

9) Subnetting Practical

Subnetting is the process of dividing a large ip network into smaller logical networks called subnets. Subnetting is perfomed due to following reasons mentioned below.

1. **Efficient IPV4 utilization**:- With the help of subnetting we can only allocate the required IP address to each department in this way we can avoid the wastage of IPv4 address.

   Example



So now we have a IP network that has 254 usable IP's (**IPv4**) excluding 1 for broadcast and 1 for network id (network address).

Required IP's are 20+50+30 = 100 IP's that means **154 IP's** are waste in 1 network and if this keeps happening in every **IP network** then the world will face shortage of ipv4 addresses.

To solve this issue we create a separate subent for each department and in this way only required IP's will be used there will be no IP's address wastage. That's why subnetting is the efficient ipv4 utilization.

2. **Improved Security**: Now lets consider if cyber attack happens on a subnett **HR** then all other subnetts are isolated means that they do not face attack consequences immediately although its on same network but hacker needs to spread his attack on entire network.

3. **Easier trouble shooting:** if there is an issue in a netowrk because of subnetting netork enginners will quickly recognize the main problem area not the entire network will be down. They will quickly identify and solve the issue.

Practice Example of IP network

192.168.1.0/24

Octet Structure of Class c IP network

192 . 168 . 1 .0/24 CIDR

N N N H

Look at the CIDR (**Classless Inter Domain Range**) its 24. this means that we have 24 **network Bits** and 8 **Host Bits.**

To calculate host bits the formula is pretty simple

Host bits = Total IP address bits - network bits

Host bits = 32 - 24

Host bits = 8

Total IP address are 2^8 = 256

To calculate total usable IP address we can use the formula

Total Usable IP address = 2^Host bits-2

Total Usable IP address = 2^8-2

Total Usable IP address = 256-2

Total Usable IP address = 254

The reason of subtracting 2 from 256 is one is Network ID and one is Broadcast ID.

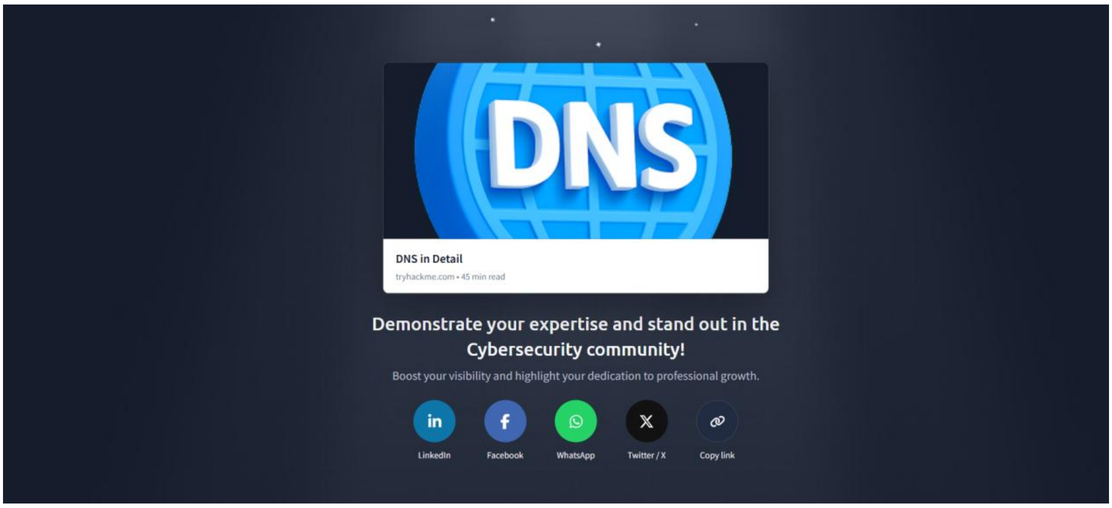
Hence this is a class c IP address therefore its default subnet Mask is **255.255.255.0**

Result

IP network = 192.168.1.0
Subnet Mask = 255.255.255.0
Total IP adrress = 256

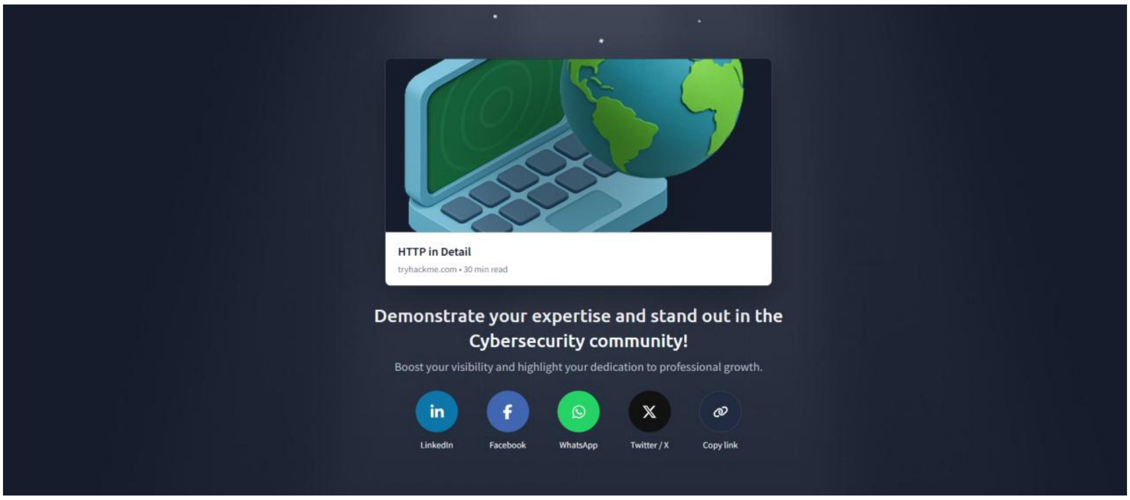
Usable IP address = 254
First Usable Host = 192.168.1.1
Last Usable Host = 192.168.1.254
Network ID = 192.168.1.0
Broadcast Address = 192.168.1.255

10) TryHackMe Room DNS in Detail



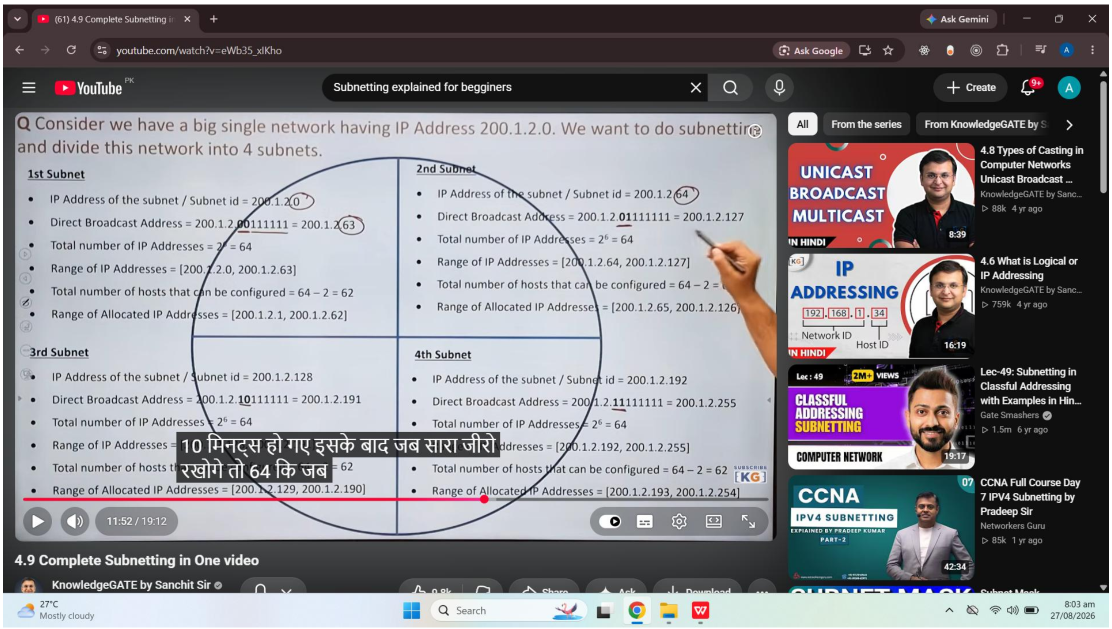
In the TryHackMe room DNS in Detail I learned about the complete working mechanism that how dns works. How domain name gets resolved into IP address. DNS record types such as A Record AAAA Record CNAME Record MX Record TXT Record. In this room I also learned about domain heirarchy like TLD (Top level Domain) Second Level Domain and Subdomain.

11) TryHackMe Room HTTP in Detail



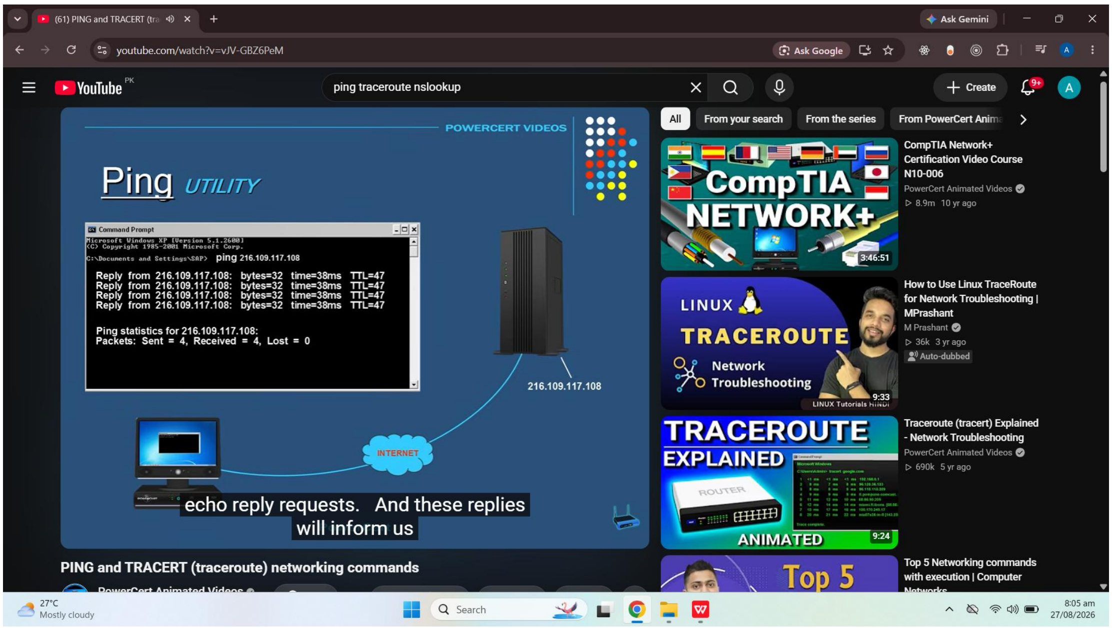
In the TryHackMe room HTTP in Detail. I learned complete details about the http protocol how it works difference between http and https.I learned how request is processed by web server and clients get the resposne back in case of any query. I also discovered the port number that http uses for the different conditions example 403 is not authorized. I dsicovered about http headers such as Content lengt content type Host name Agent. Differetn request for Data Handling like GET, POST, PUT, DELETE. Get is sued to retreive data from web server post is sue to create data in web server delete is use to delete data from web sever put is sue to update data. I also learned the importance of cookies in web servers.

12) Youtube video Subnetting explained for begginers

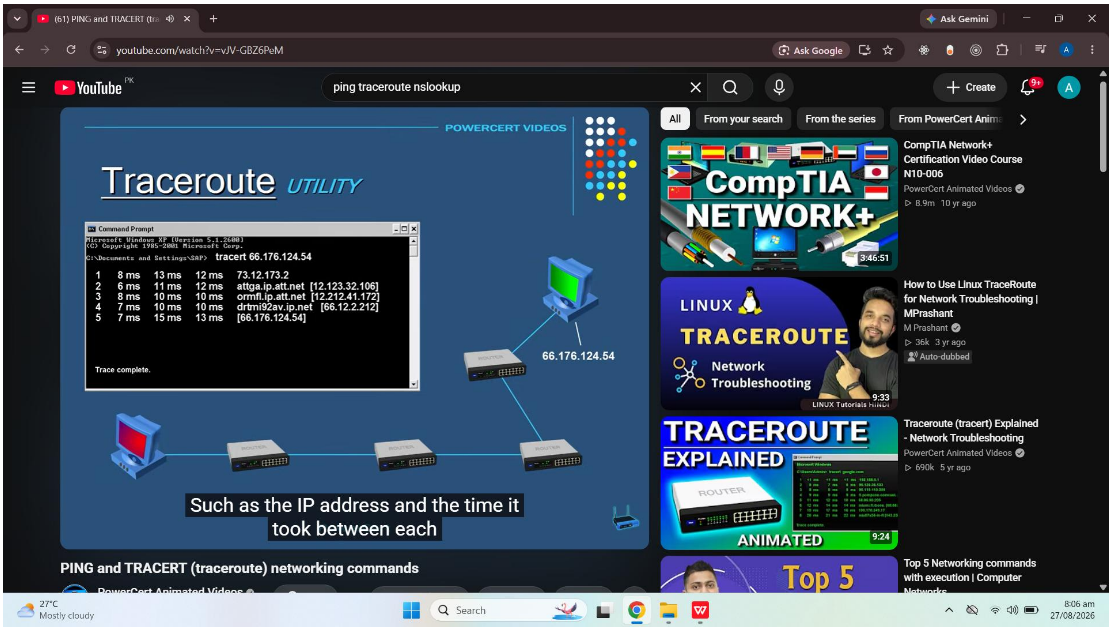


13) Youtube video explaining ping traceroute nslookup

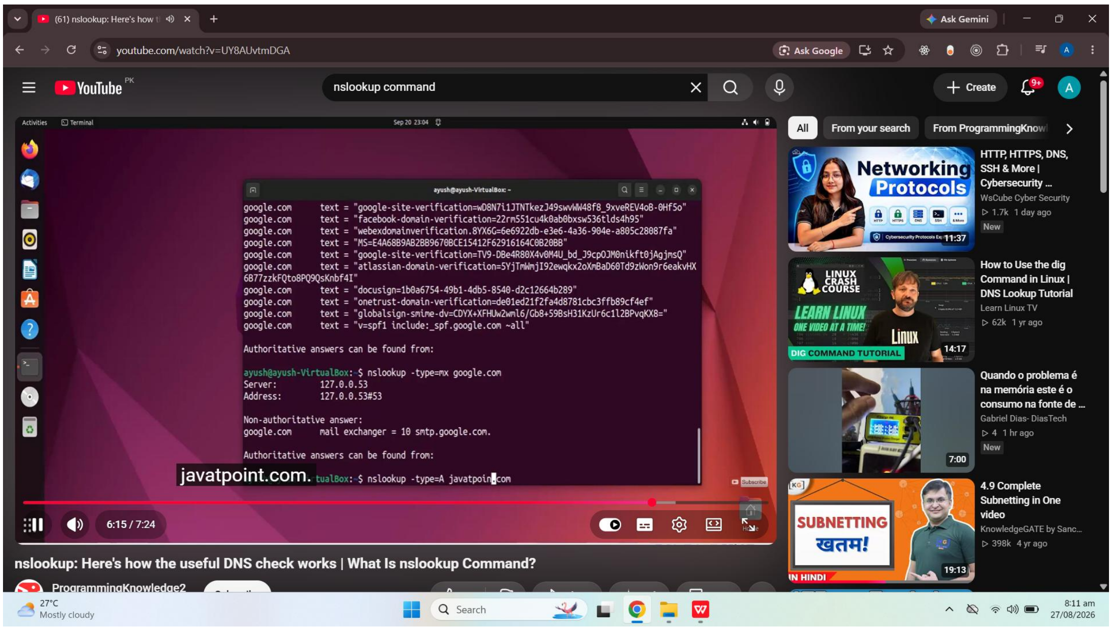
Ping command



Traceroute Command



nslookup Command



14) Conclusion

In conclusion I can say that this task helped me to understand the core networking concepts and commands use in security assessment. It also helped me to understand the importance of subnettting how subentting works understanding the cidr notation in subnetting. Other than that through this practical task I improved my concepts regarding DNS and HTTP how dns works under the hood and how http and https is different, why https is preffered, different types of http methdos, working of request and response. Over all this task enhanced my foundation for further cyber security tasks.